SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

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COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

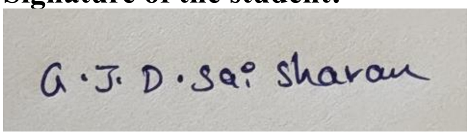
Duration: 90 minutes

Marks: 25

Code of Conduct:

I *G J D SAI SHARAN/192425152* (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Scenario-Based Questions

CO-AT2

QUESTIONS: 05 **Total Marks:25**

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. **Auto Encoders, Representation Learning**
   A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. **Width and Depth of Neural Networks, Activation Functions**
   A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. **Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**
   A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. **Deep Learning Applications, Representation Learning**

   A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established

Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Unsupervised Training of Neural Networks, Auto Encoders

Recommended Technique: Autoencoder

Justification

- ECG records are **unlabeled**, making supervised learning unsuitable.
- Autoencoders learn the **normal ECG pattern** without requiring labels.
- Abnormal ECG signals produce **high reconstruction error**, allowing anomaly detection.
- Suitable for handling large-scale medical datasets.

Workflow

1. Collect unlabeled ECG signals.
2. Preprocess the signals (noise removal and normalization).
3. Train an autoencoder using normal ECG data.
4. The encoder converts ECG signals into compact feature representations.
5. The decoder reconstructs the original ECG signal.
6. Calculate the reconstruction error.
7. If the error exceeds a threshold, classify the ECG as **abnormal**; otherwise, classify it as **normal**.

Advantages

- No labeled dataset required.
- Detects previously unseen abnormalities.
- Learns meaningful ECG representations automatically.
- Reduces manual feature engineering.

Outcome

The hospital can automatically identify abnormal ECG patterns with high efficiency and assist doctors in early diagnosis.

2. Auto Encoders, Representation Learning

Problem

Reduce the dimensionality of customer purchase data before customer segmentation.

Solution

Use an **Autoencoder** for representation learning.

Working

1. Collect customer purchase data.
2. Normalize the dataset.
3. Train an autoencoder.
4. The encoder compresses high-dimensional purchase data into a low-dimensional latent representation.
5. The decoder reconstructs the original data.
6. Use the compressed features for customer segmentation using clustering algorithms such as K-Means.

Benefits

- Removes redundant information.
- Preserves important purchasing behavior.
- Improves clustering accuracy.
- Handles high-dimensional retail data efficiently.
- Reduces computation time.

Applications

- Customer segmentation
- Personalized marketing
- Product recommendation
- Sales analysis

Outcome

The company obtains compact customer representations, leading to faster and more accurate customer segmentation.

3. Width and Depth of Neural Networks, Activation Functions

Problem

A shallow neural network performs poorly for facial recognition.

Recommended Architectural Changes

Increase Network Depth

- Replace the shallow network with a deep neural network.
- Add multiple hidden layers to learn complex facial features.
- Early layers detect edges.
- Middle layers detect facial components.
- Final layers recognize complete faces.

Increase Width

- Add more neurons in hidden layers.
- Capture richer feature representations.
- Improve learning capacity.

Activation Functions

- Replace Sigmoid with **ReLU** in hidden layers.
- Use **Leaky ReLU** if dead neuron problems occur.
- Use **Softmax** in the output layer for multi-class face classification.

Advantages

- Better feature extraction.
- Faster convergence.
- Reduced vanishing gradient problem.
- Higher recognition accuracy.

Outcome

A deeper network with ReLU activation significantly improves facial recognition performance and classification accuracy.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Problem

Build a movie recommendation system without labeled data.

Solution

Use a **Restricted Boltzmann Machine (RBM).**

Working

1. Collect users' movie viewing history.
2. Construct a user-movie interaction matrix.
3. Train the RBM using unsupervised learning.
4. Hidden neurons learn latent preferences such as genres and viewing habits.
5. The RBM predicts missing movie ratings.
6. Recommend movies with the highest predicted ratings.

Advantages

- No labeled dataset required.

- Learns hidden user preferences automatically.
- Handles sparse recommendation data.
- Improves personalized recommendations.

Applications

- Netflix
- Streaming platforms
- Music recommendation
- E-commerce recommendation

Outcome

The streaming service generates personalized movie recommendations based on users' viewing behavior.

5. Deep Learning Applications, Representation Learning

Problem

Detect defective manufacturing products using sensor readings.

Proposed Deep Learning Solution

Technique

Representation Learning using **Autoencoder + Neural Network**

Workflow

1. Collect sensor readings from manufacturing equipment.
2. Clean and normalize the sensor data.
3. Train an autoencoder to learn normal sensor behavior.
4. The encoder extracts meaningful feature representations.
5. Measure reconstruction error for each product.
6. Products with high reconstruction error are identified as defective.
7. A neural network can further classify defects into different categories if labeled data becomes available.

Advantages

- Detects defects automatically.
- Works with unlabeled sensor data.
- Learns hidden patterns efficiently.
- Supports real-time monitoring.
- Reduces inspection cost.

Applications

- Quality control
- Predictive maintenance
- Industrial automation
- Smart manufacturing

Outcome

The manufacturing company can accurately detect defective products in real time, improving product quality and reducing production losses.